

OS Semester Examination 2024

Operating Systems (OS) · Semester Examination · 2024

Max. Marks	75	Time	3 hours
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Answer ALL questions from Part A and any FIVE from Part B.

Part A (25 marks)

1. Write short notes on Operating System Structures. (5 marks)
2. Compare Scheduling Criteria with Classic Synchronization Problems. When would you prefer one over the other? (5 marks)
3. Explain Deadlock Characterization with a neat diagram and a suitable example. (5 marks)
4. Write short notes on Contiguous Allocation. (5 marks)
5. Write short notes on File Concepts. (5 marks)

Part B

1. Compare Operating System Structures with Process Concept. When would you prefer one over the other? (Unit 1)
2. Define System Calls. Derive or justify the key properties involved. (Unit 1)
3. Define Scheduling Criteria. Derive or justify the key properties involved. (Unit 2)
4. What is meant by FCFS, SJF and Round Robin? Discuss its advantages and limitations. (Unit 2)
5. Define Deadlock Characterization. Derive or justify the key properties involved. (Unit 3)
6. What is meant by Resource Allocation Graph? Discuss its advantages and limitations. (Unit 3)
7. Explain Contiguous Allocation with a neat diagram and a suitable example. (Unit 4)
8. Define Paging. Derive or justify the key properties involved. (Unit 4)
9. What is meant by File Concepts? Discuss its advantages and limitations. (Unit 5)
10. Illustrate Directory Structures with a worked example. (Unit 5)